# Dataset Report — Céré'Obs Harvest Dates

*Master Thesis: Crop Harvesting Prediction · Winter wheat · Centre-Val de Loire, France*

PREDICTION TARGET

**Dataset role:** Target variable (harvest date) · **Provider:** FranceAgriMer — Céré'Obs programme
**Source file:** SCR-GRC-CEREOBS_CP_depuis_2015-A26.xlsx · **Data cut-off:** 04/05/2026
**License:** FranceAgriMer (free public use, source attribution required)

## 1. Source and provenance

### 1.1 What is Céré'Obs?

Céré'Obs is a crop-monitoring programme run and animated by FranceAgriMer (the French national agency for agricultural and sea products). It publishes, every week from sowing to harvest, the condition and developmental progress of French cereals: soft wheat, durum wheat, winter barley, spring barley and grain maize. The observations are made on the ground by trained observers from departmental chambers of agriculture, economic organisations and technical institutes.

### 1.2 Where the data was retrieved

| Item | Detail |
|---|---|
| Access portal | FranceAgriMer VISIONet (historical time-series download service) |
| Public dashboard | cereobs.franceagrimer.fr |
| File format | Excel workbook (.xlsx), 2.86 MB |
| Temporal coverage | Harvest years 2015–2026 (weekly observations from 2014-W41) |
| Geographic scope | France, Regions, Agronomic zones (3 levels) |
| Unit of all values | % of monitored surface area |

### 1.3 Workbook structure

The file contains four sheets. The **"Données régions"** sheet was selected for this project because its granularity matches the regional study area.

| Sheet | Granularity | Use in project |
|---|---|---|
| Présentation | — | Metadata, legend, source terms (read for documentation) |
| Données France | National | Not used |
| **Données régions** | 12 regions | **Used** — matches the study area |
| Données zones agronomiques | Agronomic zones | Not used (reserved for future refinement) |

## 2. Raw data characteristics

| Property | Value |
|---|---|
| Rows (regional sheet) | 17,848 |
| Columns (non-empty) | 15 |
| Header location | Row 6 (data starts row 7) |
| Crops | Blé tendre, Blé dur, Orge d'hiver, Orge de printemps |
| Week format | ISO week string, e.g. `2023-S28` |

| | |
|---|---|
| Phenological-stage columns | Semis, Levée, Début tallage, Épi 1 cm, 2 noeuds, Épiaison, Récolte (cumulative %) |
| Crop-condition columns | Très mauvaises → Très bonnes (weekly distribution, sums to 100%) |

## 3. Transformations applied

The raw data was processed to derive a single, continuous harvest date per season. The steps below were applied in sequence.

**Step 1 — Sheet loading.** Read the "Données régions" sheet with the header positioned at row 6; dropped fully empty columns.

**Step 2 — Filtering.** Kept only rows where Culture = "Blé tendre" and Région = "Centre-Val de Loire" (540 rows across the archive).

**Step 3 — Date parsing.** Split the "Semaine" string (e.g. 2023-S28) into year and ISO-week number; mapped each to the Monday of that ISO week.

**Step 4 — Threshold crossing.** For each harvest year, located the week where cumulative "Récolte" first crosses 50% of area.

**Step 5 — Linear interpolation.** Interpolated linearly between the week below 50% and the week above to obtain a precise harvest date and day-of-year (DOY).

## 4. Key decisions and rationale

**Decision:** Use the regional sheet rather than national or agronomic-zone.
**Rationale:** The region level matches the project study area and provides one consistent harvest signal per season, avoiding both over-aggregation (national) and noise (fine agronomic zones).

**Decision:** Define the harvest date at the 50% area-harvested threshold.
**Rationale:** 50% is the most robust, unbiased indicator of the harvest peak — less sensitive to a few early or late fields than a 10% or 90% threshold. It represents the "typical" harvest moment.

**Decision:** Apply linear interpolation between weeks.
**Rationale:** Weekly data is too coarse for a timing-prediction target. Interpolation converts the discrete weekly steps into a continuous day-of-year, giving the regression model a precise target.

## 5. Resulting deliverable

### 5.1 Derived harvest dates (soft wheat, Centre-Val de Loire)

| Harvest year | Harvest date (50%) | Day of year | Note |
|---|---|---|---|
| 2020 | 3 July | 185 | Early |
| 2021 | 18 July | 199 | Latest (wet year) |
| 2022 | 2 July | 183 | Earliest (heatwave) |
| 2023 | 3 July | 184 | Early |
| 2024 | 13 July | 195 | Late |

*Inter-annual spread ≈ 15 days (earliest 2022 vs latest 2021). This year-to-year variability is precisely what the model must learn to predict from weather, soil and satellite signals.*

### 5.2 Data quality

- No missing values in key columns (Culture, Région, Semaine) or in the stage columns.
- Condition columns are sparse (blank = 0% reported) but sum to 100% whenever populated (verified on 16,728 rows).
- Stage columns are cumulative within a campaign and reset at each new crop year.

### 5.3 Documentation produced

- `Data_Dictionary_CereObs.pdf` — full column dictionary with BBCH stage mapping, value ranges, descriptive statistics and quality notes.

## 6. Relevant observations

- **Why the "Récolte" mean is only 17%:** the archive includes all weeks of the year, including autumn–spring when harvest is necessarily 0%. The average is therefore pulled down — this is expected, not a data problem.
- **Bonus predictive potential:** the earlier-stage columns (heading date, crop-condition ratings) could themselves serve as supplementary features, in addition to the weather/soil/satellite data.
- **Target alignment:** harvest dates are at regional level, so all parcels of a given year share the same target. The model therefore learns mainly to explain between-year differences, refined by parcel-level features.